Business & Functional Requirements Document

Submission deadline	Monday, 31 October 2026	Final presentation	Friday, 27 November 2026

Write this as a business manager, not a software engineer.

It should answer four questions:

what business result the application must deliver, how the work is done today and how it will be done with the app, what the app must do for each user, and how your team will get it built, tested and live by 27 November. The AI app-builder will handle the code; the clarity of this document decides the quality of what it builds. Maximum five pages, plus an appendix for diagrams. Replace every grey prompt with your own words and submit as PDF named **S<Section>_G<Group>_BRD.pdf**.

A TEAM AND PROJECT DETAILS

Section	1	Group no.	3
Group name	ASML	Team lead (email)	antheranjit.dm281004@greatlakes.edu.in
Application name	**StockSaathi** (working title)		
Changes since Topic Brief	**1. Forecast method.** The Topic Brief used one formula (14-day average). We now use a Forecast Desk: five simple models, each checked against recent real sales and blended by accuracy, the way weather centres combine several forecast models. Reason: kirana demand moves with weekdays, salary week and festivals, and one average reacts late. The old formula stays as one of the five models and as the fallback. **2. Stock accuracy.** Added 'record a delivery' so stock stays correct. Added an order-cover setting so daily items such as milk are not ordered a week at a time. **3. Login and language.** Google sign-in instead of phone OTP (OTP needs a paid SMS provider). English only by 27 November; a regional language after the pilot. **Faculty feedback:** none received yet on the Topic Brief; the changes above are our own.		

1 BUSINESS REQUIREMENTS *Why the business needs this application.*

Business problem	Kirana owners (about 13 million stores in India; Cornell, 2026) reorder hundreds of items from memory, a notebook or the distributor salesman's advice. No sales history is used. Fast items run out and slow items lock up cash and expire. Retail out-of-stock rates average 8.3%. When an item is missing, 31% of shoppers buy it at another store and 9% do not buy it at all (Gruen, Corsten & Bharadwaj, 2002). Quick-commerce apps now deliver in about 10 minutes, so a missing item can lose the customer to an app. **Local evidence (simulated, to be replaced by our interview at a general store near the GLIM campus, Manamai, by 6 Nov):** about 6 stockouts a week on top items (milk, bread, instant noodles, biscuits, water); about ₹1,500 of expired stock thrown away each month; about 25 minutes to prepare each distributor order; 3 distributor salesmen visit each week.
Business objectives	**O1.** Cut the out-of-stock rate on the pilot store's top 20 items from the week-1 baseline to below 4% within 12 months of launch. **O2.** Cut the owner's time to prepare one distributor order to under 5 minutes within 3 months of launch. **O3.** Cut the value of slow-moving stock (more than 30 days of cover) by 25% within 6 months of launch.
Who pays, and why	The store owner pays ₹99 a month (about ₹3.30 a day) after a 1-month free trial. Today's method costs no money, but it costs lost sales, expired stock and the owner's time every evening. The fee is worth paying if the app saves a few lost sales a week. We will test this price with 5 owners before launch. Later option: an FMCG distributor pays for its retailers, in return for cleaner and earlier order data.

Business outcomes and success measures *How you will prove the app made a difference. 3–4 rows.*

Business outcome	Measure (KPI)	Today (baseline)	Target	How we will measure it
Fewer lost sales from empty shelves	Out-of-stock rate on top 20 items (share of item-days with zero stock)	Measured at pilot store in week 1 (global average 8.3%)	Below 4% in 12 months	Daily shelf check in week 1, then app stock records
Faster ordering	Owner's minutes to prepare one distributor order	About 25 minutes (simulated interview); timed again in week 1	Under 5 minutes in 3 months	App time from opening the list to tapping Share
More accurate forecasts	Average units missed per item per day, top 20 items	Error of the simple 14-day average (Topic Brief formula)	At least 10% lower error over the 4-week test	App compares each forecast with real sales (FR10)
Less cash stuck in slow stock	₹ value of items with more than 30 days of cover	Measured in week 1	25% lower in 6 months	Don't reorder report (FR6) × item cost

2 STAKEHOLDERS AND ACCESS *Everyone affected by the app, and what each is allowed to do in it.*

Stakeholder	Role in the business process	What they need from the app	Access level	Approves / signs off
Store owner	Runs the shop; decides and places reorders	A ready reorder list each evening, with quantities and a reason	Administrator (own shop: items, helpers, settings)	Yes: own orders, stock and prices
Helper or family member	Serves at the counter	One-tap sale entry that does not slow the queue	Full user (records sales only)	No
Distributor salesman	Takes and delivers orders	A clear order on WhatsApp; fewer last-minute calls	No login (receives messages)	No; delivers against the order
Shoppers	Buy from the store	Items on the shelf when they come	No access (no shopper data stored)	No
Group 3 product team	Builds and supports the pilot	Error logs and forecast accuracy reports	Administrator (app settings and festival calendar; cannot change shop records)	Signs off each release

Access levels: No access (affected but does not use the app) · View only · Full user (uses the main features for their own records) · Manager (sees and approves others' records) · Administrator (manages users and settings).

3 BUSINESS PROCESS DEFINITION *How the work happens today, and how it will happen with the app.*

Current process (As-Is) *4–7 steps. Mark where time, money or quality is lost.*

#	Who	What happens today	Pain point (delay, cost, error, risk)
1	Shopper and owner	A shopper asks for an item and the shelf is empty.	Lost sale. 31% buy elsewhere, 9% buy nothing (Gruen et al., 2002).
2	Helper	Sells all day. Nothing is written down per item. UPI shows amounts, not items.	No item-level sales history, so no basis for reordering.
3	Store owner	Checks shelves by eye after closing and writes a list in a notebook.	About 25 minutes per order (simulated interview); fast items are missed.
4	Distributor salesman	Visits once a week and suggests what to order.	Owner orders what the salesman pushes; slow stock builds up and expires.
5	Store owner	Phones the order or tells the salesman; checks stock when it arrives.	Wrong quantities, repeat calls, cash locked in the wrong items.

Future process with the app (To-Be) *Same process, redesigned. Show which steps the app removes, speeds up or automates.*

#	Who	What happens with the app	Business improvement
1	Helper or owner	Taps the item sold (one tap), or uploads past sales as a CSV file on day one.	**Speeds up:** item-level sales history from day one, under 10 seconds a sale.
2	App (every night)	Forecast Desk runs five models per item, checks each against the last 14 days of real sales, and blends them into one forecast with a confidence level.	**Automates:** guesswork replaced by a forecast that adjusts for weekdays, salary week and festivals.
3	App	Builds the 'Reorder today' list with quantities and the 'Don't reorder' list.	**Removes:** the evening shelf check. Slow stock is flagged before it is reordered.
4	Store owner	Reviews the list, edits any quantity and taps Share order.	**Speeds up:** order ready in under 5 minutes.
5	Distributor, then owner	Salesman gets the order on WhatsApp and delivers. Owner records the delivery; stock updates.	**Improves:** stock stays correct and tomorrow's forecast uses today's real sales.

Process diagram	To-Be swim-lane diagram (lanes: Helper, Store owner, StockSaathi app, Distributor salesman; decisions: slow or overstocked?, stock at or below reorder point?, change any quantity?): see **Appendix A1**.

4 FUNCTIONAL REQUIREMENTS *What the app must do, in plain business language. 6–10 requirements.*

ID	The app must allow [user] to [do something]	Process step	Priority	We will know it works when…
FR1	The app must allow the helper or owner to record a sale in one tap from a list of items.	To-Be 1	Must	A sale is saved in under 10 seconds and appears in today's sales. With no network, it is saved on the phone and sent later.
FR2	The app must allow the owner to upload past sales as a CSV file (date, item, quantity).	To-Be 1	Must	The 30-day demo file (1,633 rows) loads in under 1 minute; any bad rows are listed, not lost.
FR3	The app must allow the owner to set up items (price, distributor, lead time, minimum safety stock, order cover days) and record deliveries.	To-Be 5	Must	After a delivery, stock = old stock + delivered − sold, and matches a hand count.

ID	The app must allow [user] to [do something]	Process step	Priority	We will know it works when…
FR4	The app must produce each night a demand forecast for every item from five models, check each model against the last 14 days of real sales, and blend them by accuracy (Forecast Desk, Appendix A4).	To-Be 2	Must	For a test item, the blended forecast and weights match a hand calculation from the five model numbers and their past errors (worked example in A4).
FR5	The app must show the owner a 'Reorder today' list with quantities and a confidence level.	To-Be 3	Must	With demo data, every item at or below its reorder point is listed and each quantity matches BR1 and BR2.
FR6	The app must show the owner a 'Don't reorder' list of slow and overstocked items.	To-Be 3	Must	All 3 planted slow items in the demo data are flagged and none is on the reorder list.
FR7	The app must allow the owner to edit quantities and share the order with the distributor on WhatsApp.	To-Be 4	Must	Share opens WhatsApp with the distributor's number and the order text; the order is saved as Sent.
FR8	The app must allow the product team to enter festival and holiday dates for the look-alike model.	To-Be 2	Must	A date marked as a festival changes that week's look-alike forecast.
FR9	The app must show the owner 'Why this number' for an item: each model's forecast, its weight and any condition.	To-Be 3	Nice	An owner explains a reorder quantity to us without help.
FR10	The app must show a weekly scorecard: out-of-stock days, forecast error against the simple average, cash in slow stock.	KPIs	Nice	Scorecard numbers match the same numbers worked out in Excel from the raw data.

Priority: Must = the app is useless without it and it will be built by 13 November. Nice = only if time allows.

Business rules *The policies the app must enforce, e.g. limits, approvals, eligibility, pricing, deadlines.*

BR1	Reorder: an item goes on 'Reorder today' when stock plus anything already ordered is at or below its reorder point. Reorder point = forecast daily demand × lead time (days) + safety stock.
BR2	Order quantity = forecast daily demand × (lead time + order cover days) + safety stock − stock − already ordered, rounded up to whole units. Order cover is 7 days by default and 1 day for daily-delivered items such as milk and bread.
BR3	Safety stock = the larger of the owner's minimum and the models' disagreement (highest minus lowest model forecast) × lead time, capped at 7 days of demand.
BR4	Don't reorder (overrides BR1): no sale in the last 14 days, or more than 30 days of stock in hand.
BR5	Model weights are reset every night from each model's average miss over the last 14 days: a smaller miss gives a bigger weight, with no model below 5% or above 60%. New items (under 14 days of sales) and slow items (under 1 sale a day) use only the 28-day average and show 'Low confidence'. Confidence is High when the models' spread is within 20% of the forecast, Medium within 50%, Low above that.
BR6	Guard rail: if the blended forecast is more than double the 28-day average and no festival or event is marked, the app caps it at double and flags it for the owner to check.
BR7	Only the owner can send orders or change stock, prices and settings; helpers can only record sales. No shopper personal data is recorded.

Out of scope	Payments and UPI; distributor login; connection to billing software (simulated by CSV upload); barcode scanning; reading bill photos; regional language; more than one shop per owner; machine-learning model training (the Forecast Desk uses five rule-based statistical models); live rain-forecast input; phone OTP login.

5 USER FLOW *The path each type of user takes through the app, screen by screen.*

User flow diagram	One flow per user type, screen by screen: see **Appendix A2**.
Flow in words	**Store owner (daily):** Signs in with Google → Home (Reorder today: 6, Don't reorder: 3) → taps Reorder list → (optional) taps an item → Why this number → back → edits quantities → taps Share order → WhatsApp opens with the order → sends → back to Home. On delivery day: Home → taps Deliveries → enters quantities → taps Save. **Helper (during the day):** Quick sale screen (top items as large buttons) → taps item, adjusts quantity → Saved tick → next customer. **Owner set-up (first day):** Add items or Upload CSV → upload result (rows saved, errors) → sets distributor, lead time and minimum stock → items ready. **Distributor salesman:** no screens; receives a WhatsApp message with items and quantities.

6 HOW IT WILL BE BUILT *Decisions only. The AI app-builder does the technical work.*

AI app-builder	☐ Lovable ☐ Bolt ☐ Replit ☐ v0 ☒ Other: Claude Code (Next.js); fallback Lovable if not approved *Free tier only.*
Where the data is kept	☐ Builder's built-in database ☒ Supabase ☐ Firebase ☐ Google Sheets ☐ No data saved
How users log in	☐ No login ☐ Email & password ☒ Google sign-in ☐ Phone OTP

Outside services used	☐ None ☐ Maps ☒ WhatsApp / SMS link ☐ Payments (test) ☐ AI feature ☒ Other: Vercel daily scheduled job for the nightly forecast
Where it will go live	☐ Builder's Publish link ☒ Vercel ☐ Netlify ☐ Other: _____
Application structure	**Picture:** Owner and helper phones → App screens (Next.js on Vercel) → App logic (sales, reorder rules, Forecast Desk) → Supabase database (12 linked tables) and file storage → WhatsApp link to the distributor. A Vercel daily job runs the Forecast Desk each night (the free plan allows one run a day). See **Appendix A3** for the structure and how the tables connect. **Existing systems:** the owner's notebook or billing software is simulated by CSV upload. UPI apps are not connected (they record amounts, not items).

7 DATA AND PRIVACY: FIRST VIEW

Personal data collected	**Owner:** name and Google email, to log in and to own the shop's records. **Helper:** name and Google email, to log in and to show who recorded each sale. **Distributor salesman:** name and WhatsApp number, entered by the owner with the salesman's agreement, to send orders. **Not collected:** shopper details, location tracking, photos, payment or health data. **Who sees it:** only that shop's owner (database rule per shop); helpers see item names only; the product team sees error logs without shop records. A consent notice is shown at sign-up (DPDP Act, 2023), and the owner can delete the account and all data.

8 MILESTONE PLAN *Fixed deadlines. Fill in what you will deliver and who leads each milestone.*

Milestone	Deadline	What we will deliver	Lead(s)
1. Ideation	Completed	*Topic Brief approved.*	Aditya Ranjith, Shivang Singh
2. Requirements	Mon, 31 Oct 2026	This document, with the To-Be process and user-flow diagrams (Appendix).	Aditya Ranjith, Shivang Singh, Mitansh Sharma, Yousuf Khan, Niveada N
3. Development	Fri, 13 Nov 2026	Stage A by Fri 6 Nov: FR1–FR3 and FR5–FR7 working on the simple 14-day average. Stage B by Fri 13 Nov: Forecast Desk (FR4, FR8). Prompt history saved.	Eesha Meenakshi, Shruti Nair, Samruddhi Patil (build); Apeksha Singla, Harshit Sharma, Subhangi Samal (data)
4. Testing	Fri, 20 Nov 2026	Every FR checked against its test; forecast back-test on the demo data; at least 5 real users; data-privacy check; issues fixed.	Shaurya Singhal, Amish Siddiqui, Vaishnavi Kolhe, Radha Nath Singh
5. Deployment	Tue, 24 Nov 2026	Live on a Vercel link that works on a low-cost Android phone, loaded with demo data (30 items, 1,633 sales, 4 distributors).	Apeksha Singla, Harshit Sharma, Subhangi Samal
6. Final submission	Fri, 27 Nov 2026	*Presentation on the application and the process followed: live demo, business outcomes, lessons learnt.*	Aditya Ranjith, Shivang Singh, Mitansh Sharma, Yousuf Khan, Niveada N

Testing approach	**Who:** all 15 team members run the test for each FR on their own phones (Android and iPhone), plus at least 5 real users: 2–3 kirana owners near campus and their helpers. Each real user logs 10 sales, reads the reorder list and shares one order without help. One low-cost Android phone is tested on a slow connection. **Forecast back-test:** run the Forecast Desk on the first 16 days of demo data and compare with the last 14 days. **Ready to go live when:** every Must FR passes its test; 5 users finish the three tasks unaided; a second shop account cannot see the first shop's data; and the Forecast Desk is no worse than the simple average (if it is worse, the app uses the simple average).

9 ASSUMPTIONS AND RISKS

Key business assumptions	1. Owners or helpers will log sales on at least 6 of 7 days (check: app logging rate in pilot week 1). 2. Owners will act on a suggested list (check: share of suggested quantities sent unchanged). 3. Distributors accept orders on WhatsApp (check: ask 2 salesmen during interviews). 4. Owners will pay ₹99 a month (check: price question to 5 owners). 5. Two to four weeks of sales is enough for the models to beat a simple average (check: back-test; fallback in BR5).
Top 3 risks	**1. Adoption:** helpers skip logging at busy times. Action: start with the top 20 items, large one-tap buttons sorted by today's sales, CSV upload as a back-up, and measure logging effort in week 1. **2. Forecast Desk takes longer to build:** Action: Stage A ships on the simple average by 6 Nov; if the full Desk is not ready by 13 Nov, demo it with three models (M1–M3). **3. Free-tier limits:** the Vercel free plan runs a scheduled job once a day (start time ±59 min), and free Supabase projects pause after a week without use. Action: one nightly run is enough; the forecast also runs when the owner opens the app; the team opens the project daily during the pilot. Data size is about 22 MB per shop per year against a 500 MB limit.

Question for faculty	Is Claude Code (Next.js, Supabase, Vercel, free tiers) approved as our app-builder, as asked in the Topic Brief? If not, we will build with Lovable on the same Supabase database.

10 EVIDENCE OF TEAM WORK AND AI USE

Links (view access)	**Miro board:** To-Be process and user flows (exported as Appendix A1 and A2). **Interview notes:** campus general store, Manamai (simulated figures in Section 1; real notes after the visit). **Prompt history:** docs/prompt-history.md in the StockSaathi code repository. **Data workbook:** StockSaathi_Data_Workbook.xlsx (Session 6). **Task board:** milestone plan in Section 8.
How AI was used	Claude (Anthropic) drafted this document, the Forecast Desk design, the appendix diagrams and the first app code from our Topic Brief and Session 6 data workbook. **Team checks:** we checked every figure against its source and the data workbook; changed the login to Google sign-in after finding that phone OTP needs a paid SMS provider; added order-cover days so milk is not ordered a week at a time; and rejected a single-formula forecast in favour of the multi-model Forecast Desk.
Who did what	**Requirements & process:** Aditya Ranjith, Shivang Singh, Mitansh Sharma, Yousuf Khan, Niveada N **Design & user flow:** Shaurya Singhal, Amish Siddiqui **Build (prompts):** Eesha Meenakshi, Shruti Nair, Samruddhi Patil, Apeksha Singla, Harshit Sharma, Subhangi Samal **Testing:** Vaishnavi Kolhe, Radha Nath Singh, Shaurya Singhal, Amish Siddiqui **Presentation:** Aditya Ranjith, Shivang Singh, Mitansh Sharma

Sources: Cornell SC Johnson College of Business, 'India's digital pull revolution' (2026); Gruen, Corsten & Bharadwaj, Retail Out-of-Stocks (GMA, 2002); Krishnamurti et al., 'Improved weather and seasonal climate forecasts from multimodel superensemble', Science 285 (1999); Makridakis, Spiliotis & Assimakopoulos, 'The M4 Competition', Int. J. Forecasting 34(4) (2018); Vercel and Supabase plan documentation (Oct 2026).

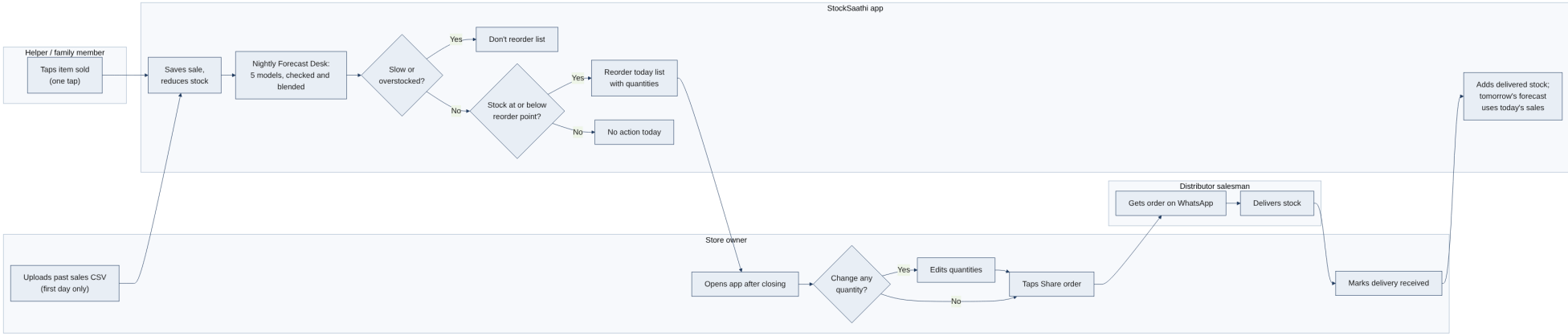
☐ We confirm that this document reflects our team's own business analysis and decisions, that all members contributed, and that we can explain every part of it.

FOR FACULTY USE

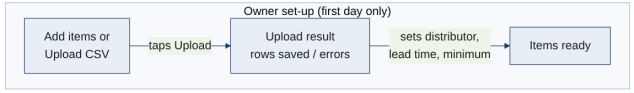
Decision	☐ Approved to build ☐ Approved with changes ☐ Revise and resubmit
Comments	

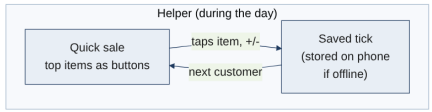
APPENDIX *Diagrams referred to in the document.*

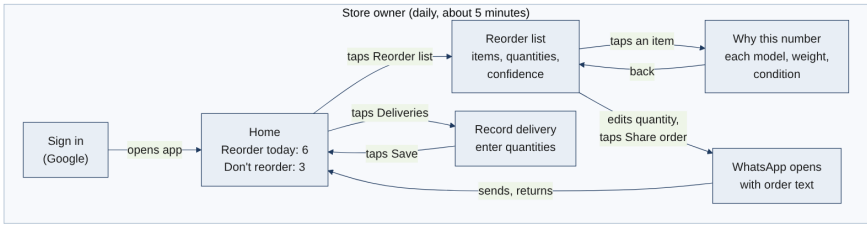
A1 To-Be business process (swim lanes) *Decision points shown as diamonds. Supports Section 3.*



A2 User flows, one per user type *Screens as boxes; arrows show what the user does. Supports Section 5.*

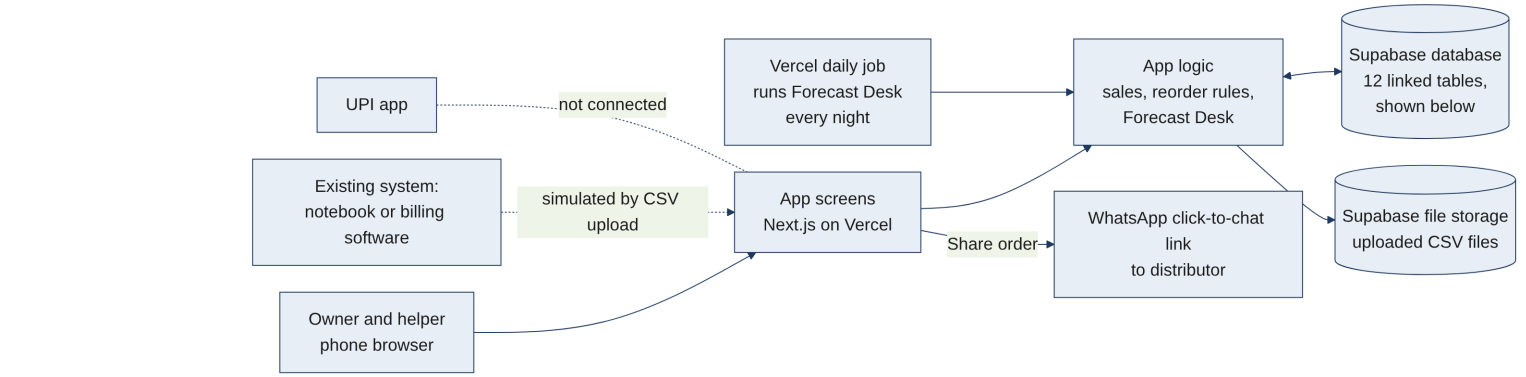


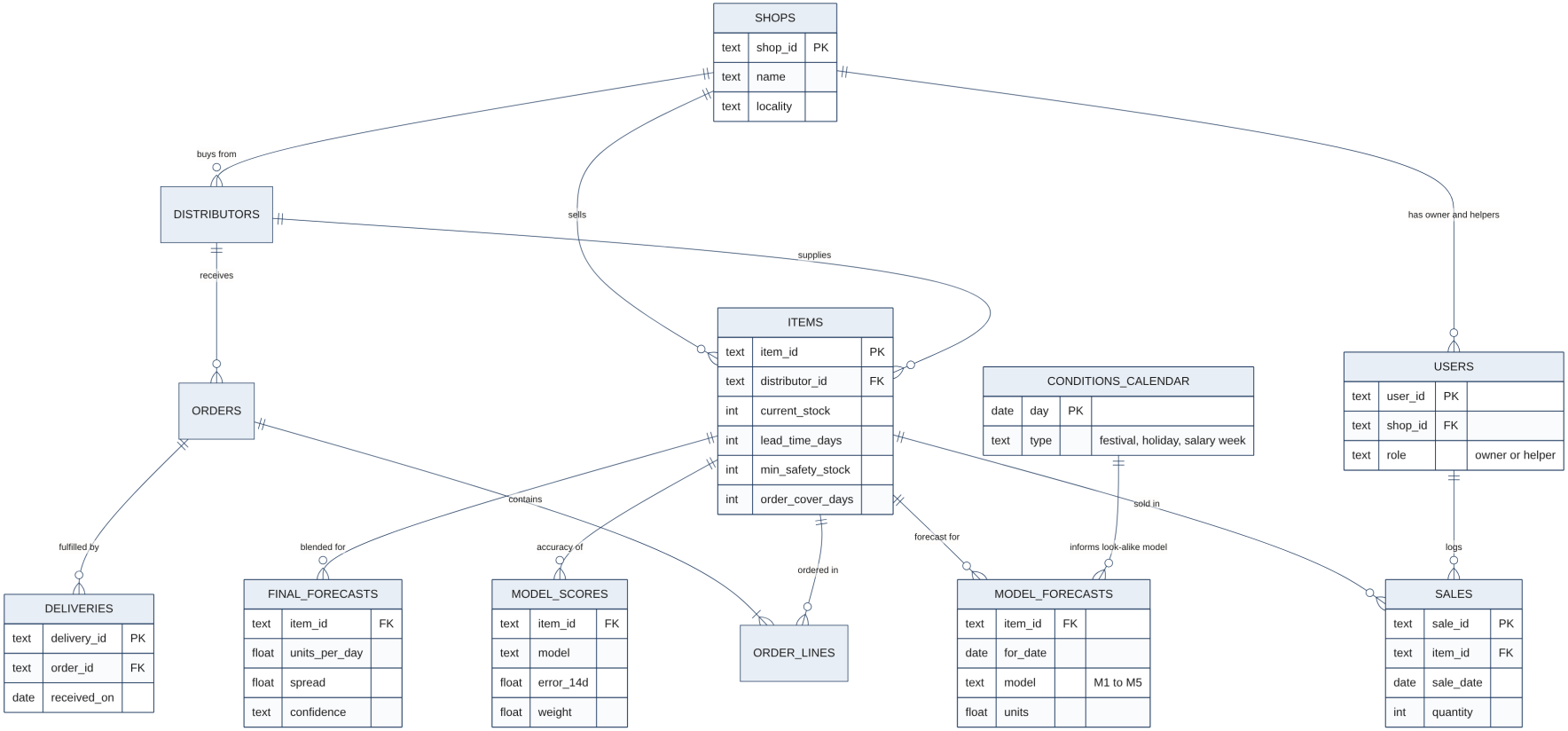




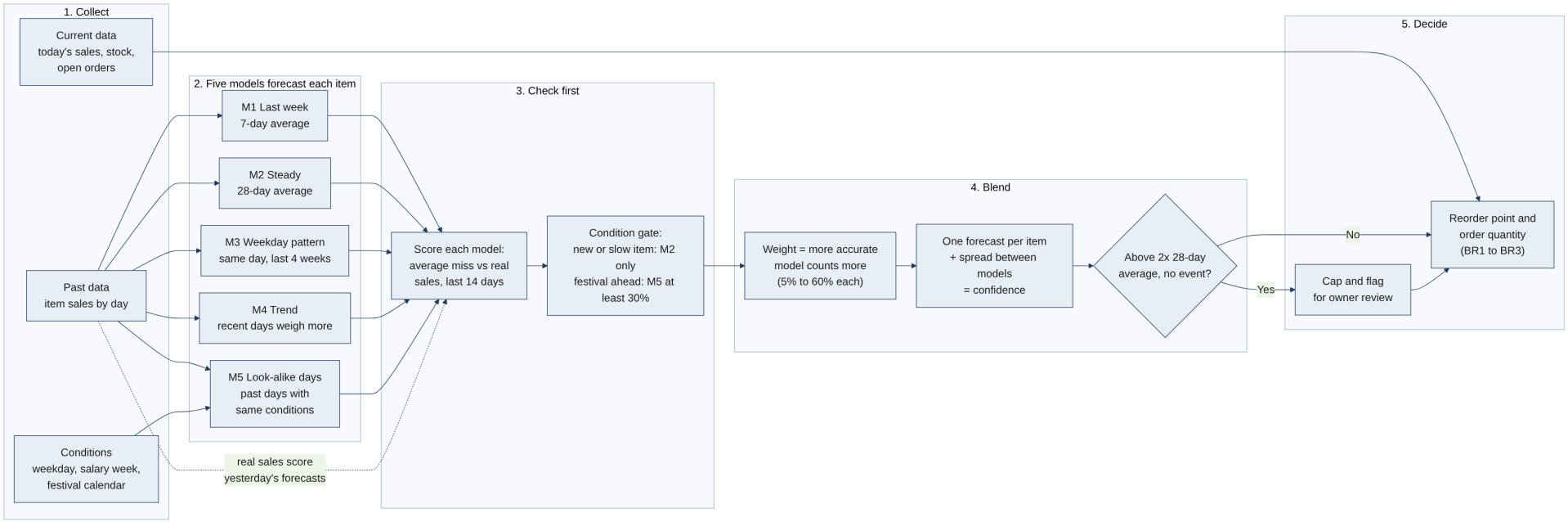
A3 Application structure and how the data connects

Top: the parts of the app. Bottom: the 12 database tables and their links. Supports Sections 6 and 7.





A4 Forecast Desk: how the multi-model forecast works *Supports FR4 and BR1 to BR6.*



Worked example (hand check for FR4): Sugar 1 kg, lead time 2 days, owner's minimum safety stock 4, stock 12, nothing on order

Model	What it looks at	Forecast (units/day)	Average miss, last 14 days	1 ÷ miss	Weight
M1 Last week	Average of the last 7 days	9.6	2.0	0.500	21.4%
M2 Steady	Average of the last 28 days (Topic Brief formula)	8.7	2.5	0.400	17.1%
M3 Weekday pattern	Same weekday, last 4 weeks	9.0	2.0	0.500	21.4%
M4 Trend	Recent days count more (smoothing)	9.4	1.6	0.625	26.7%
M5 Look-alike days	Past days with the same conditions (salary week)	11.0	3.2	0.313	13.4%
Blend	Weighted average of M1 to M5	9.45		2.338	100%

Then: spread = 11.0 − 8.7 = 2.3 units (24% of 9.45, so Medium confidence). Safety stock = larger of 4 and 2.3 × 2 = 4.6. Reorder point = 9.45 × 2 + 4.6 = 23.5. Stock 12 is below 23.5, so sugar goes on 'Reorder today'. Order = 9.45 × (2 + 7) + 4.6 − 12 = 77.7, rounded up to **78 units**. Figures are illustrative.